02 — The Feynman–Kac Formula

Goal of this note. Prove that a conditional expectation of a functional of an SDE solves a linear parabolic PDE, and conversely. This is the *linear, uncontrolled* version of everything that follows:

This note	Later
$u(t,x) = \mathbb{E}[\,\cdot \mid X_t = x]$	value function $v(t,x) = \sup_\alpha \mathbb{E}[\,\cdot\,]$ (note 03)
linear PDE $\partial_t u + \mathcal{L} u = 0$	HJB $\partial_t v + \sup_\alpha \mathcal{L}^\alpha v = 0$ (note 04)
proof via Itô + martingale	verification theorem (note 04)
linear: $u \leftrightarrow \mathbb{E}$	nonlinear: $v \leftrightarrow$ BSDE (note 08)

If this note is clear, the HJB derivation in note 04 is the same argument with one "sup" added.

Prerequisites. Note 01: Itô's lemma (5.1), the jump version (6.3), the generator (7.1), Dynkin's formula (7.2).

Notation

Symbol	Meaning	Type
X_s, $s \in [t,T]$	state process, started at $X_t = x$	real-valued process
$b(s,x)$, $\sigma(s,x)$	drift and volatility of X	functions, Lipschitz in x
$\mathcal{L}$	generator: $\mathcal{L}\varphi = b\,\partial_x\varphi + \frac{1}{2}\sigma^2\partial_{xx}\varphi$ (+ jump term in §5)	operator
$g(x)$	terminal payoff	function
$f(s,x)$	running payoff rate (paid per unit time)	function
$r(s,x) \geq 0$	discount (killing) rate	function
$D_{t,s} = \exp\big(-\int_t^s r(\tau, X_\tau)\,d\tau\big)$	discount factor from t to s	process
$u(t,x)$	solution of the PDE / the expectation	function
$\tau = T - t$	time to maturity	≥ 0
$\mathbb{E}_{t,x}[\cdot]$	shorthand for $\mathbb{E}[\cdot \mid X_t = x]$	—

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1. Statement

Theorem 1.1 (Feynman–Kac). Consider the SDE

$$dX_s = b(s,X_s)\,ds + \sigma(s,X_s)\,dW_s, \qquad X_t = x, \tag{1.1}$$

and the terminal-value problem

$$\partial_t u + b\,\partial_x u + \tfrac{1}{2}\sigma^2 \partial_{xx} u - r\,u + f = 0 \quad \text{on } [0,T)\times\mathbb{R}, \qquad u(T,x) = g(x). \tag{1.2}$$

Assume: - **(A1)** b, σ are Lipschitz in x with linear growth, so (1.1) has a unique solution with finite moments of all orders; - **(A2)** $r \geq 0$ is bounded below, and f, g have polynomial growth; - **(A3)** $u \in C^{1,2}$ solves (1.2) and has polynomial growth in x.

Then

$$\boxed{u(t,x) = \mathbb{E}_{t,x}\Big[D_{t,T}\, g(X_T) + \int_t^T D_{t,s}\, f(s, X_s)\, ds\Big].} \tag{1.3}$$

How to read it. $u(t,x)$ is the present value, at time t in state x, of: - a terminal payoff g received at T, - plus a cash flow paid at rate f over $[t,T]$, - all discounted at rate r.

The PDE (1.2) is solved **backward** from the terminal condition at T, matching the fact that payoffs are known at maturity and values are needed today.

> **Intuition for a rates quant:** this is exactly why a bond or swap price satisfies a pricing PDE. With X the short rate, $r(s,x) = x$, $g = 1$, and $f = 0$, (1.3) is the zero-coupon bond price $\mathbb{E}[e^{-\int r}]$ and (1.2) is the term-structure PDE you already use.

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2. Proof: PDE $\Longrightarrow$ expectation

Roadmap. Build a process M_s from u that (i) equals $u(t,x)$ at $s = t$, (ii) equals the quantity inside the expectation in (1.3) at $s = T$, and (iii) has zero drift by the PDE. Then M is a martingale and $\mathbb{E}[M_T] = M_t$ is exactly (1.3).

Step 1: define the candidate martingale. For $s \in [t,T]$, let

$$M_s = D_{t,s}\, u(s, X_s) + \int_t^s D_{t,\tau}\, f(\tau, X_\tau)\, d\tau. \tag{2.1}$$

Check the endpoints: - At $s = t$: $D_{t,t} = 1$ and the integral is empty, so $M_t = u(t,x)$. - At $s = T$: $u(T, X_T) = g(X_T)$, so M_T is the expression inside the expectation in (1.3).

Step 2: differentiate the discount factor. $D_{t,s}$ is an ordinary (pathwise differentiable) function of s, because its exponent is an ordinary $d\tau$-integral:

$$dD_{t,s} = -r(s, X_s)\, D_{t,s}\, ds. \tag{2.2}$$

It has no dW part, so its covariation with anything is zero.

Step 3: Itô on $u(s, X_s)$. By note 01 eq. (5.1),

$$du(s, X_s) = \big(\partial_s u + \mathcal{L} u\big) ds + \sigma\, \partial_x u\, dW_s. \tag{2.3}$$

Step 4: product rule. By note 01 eq. (5.3), $d(D\,u) = D\,du + u\,dD + dD\,du$. The covariation $dD\,du$ is zero (Step 2). Using (2.2)–(2.3):

$$d(D_{t,s} u) = D_{t,s}\big(\partial_s u + \mathcal{L} u - r\,u\big) ds + D_{t,s}\,\sigma\,\partial_x u\, dW_s.$$

Step 5: add the running payoff. The integral term in (2.1) contributes $D_{t,s} f \, ds$. So

$$dM_s = D_{t,s} \underbrace{\big(\partial_s u + \mathcal{L} u - r\, u + f\big)}_{=\,0 \text{ by } (1.2)} ds + D_{t,s}\, \sigma\, \partial_x u \, dW_s = D_{t,s}\, \sigma\, \partial_x u \, dW_s. \tag{2.4}$$

Step 6: the stochastic integral is a true martingale. By note 01 §3, we need $\mathbb{E}\int_t^T D^2 \sigma^2 (\partial_x u)^2 ds < \infty$. Under (A1)–(A3), $D \le e^{c(T-t)}$ (with $-c$ the lower bound of r; $c = 0$ when $r \ge 0$), σ^2 has quadratic growth, $\partial_x u$ is assumed to have polynomial growth (a standard part of (A3); it holds for all examples below), and X has finite moments of all orders. So the integrand is integrable.

> **Remark:** this is the step where the growth assumptions are actually used. Without them, the integral is only a local martingale and (1.3) can fail (see Pitfall 2).

Step 7: take expectations. A martingale has constant expectation, so $\mathbb{E}_{t,x}[M_T] = M_t = u(t,x)$, which is (1.3). □

Recap. The PDE is precisely the condition that the "discounted value plus accumulated cash flows" has no drift. Pricing = finding the function that makes this process a martingale.

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3. Converse (heuristic): expectation ⟹ PDE

Roadmap. Now go the other way: *define* u by the expectation (1.3) and derive the PDE. The argument below is formal (it assumes u is smooth), but it is exactly the dynamic programming argument of note 03 with the "sup" removed, so it is worth seeing carefully.

For simplicity take $r = 0$ and $f = 0$, so $u(t,x) = \mathbb{E}_{t,x}[g(X_T)]$.

Step 1: tower property + Markov property. Fix a small $h > 0$. Conditioning first on the information at $t + h$:

$$u(t,x) = \mathbb{E}_{t,x}\Big[\mathbb{E}\big[g(X_T) \mid \mathcal{F}_{t+h}\big]\Big].$$

Because X solves an SDE, it is Markov: the future after $t+h$ depends on the past only through X_{t+h}. Hence $\mathbb{E}[g(X_T) \mid \mathcal{F}_{t+h}] = u(t+h, X_{t+h})$ and

$$u(t,x) = \mathbb{E}_{t,x}\big[u(t+h, X_{t+h})\big]. \tag{3.1}$$

In words: today's value is the expected value tomorrow. **This is the dynamic programming principle without control.**

Step 2: expand with Dynkin. Apply note 01 eq. (7.2) to u on $[t, t+h]$:

$$\mathbb{E}_{t,x}\big[u(t+h, X_{t+h})\big] = u(t,x) + \mathbb{E}_{t,x}\Big[\int_t^{t+h} (\partial_s + \mathcal{L}) u(s, X_s)\, ds\Big].$$

Step 3: combine and shrink h. Substituting into (3.1), the $u(t,x)$ terms cancel:

$$0 = \mathbb{E}_{t,x}\Big[\int_t^{t+h} (\partial_s + \mathcal{L})u(s, X_s)\, ds\Big].$$

Divide by h and let $h \to 0$. By continuity of the integrand and of paths, the average over $[t, t+h]$ converges to its value at (t,x):

$$(\partial_t + \mathcal{L})u(t,x) = 0. \tag{3.2}$$

Also $u(T,x) = \mathbb{E}_{T,x}[g(X_T)] = g(x)$. With r, f restored, the same steps give (1.2).

Recap. Expectation $\Longrightarrow$ (DPP) $\Longrightarrow$ (Itô/Dynkin) $\Longrightarrow$ PDE. In note 04, (3.1) becomes $v(t,x) = \sup_\alpha \mathbb{E}[v(t+h, X_{t+h})]$ and (3.2) becomes $\sup_\alpha(\partial_t + \mathcal{L}^\alpha)v = 0$.

> **What makes it rigorous:** the gap is assuming $u \in C^{1,2}$. For uniformly parabolic problems ($\sigma^2 \geq \epsilon > 0$) with smooth data, parabolic regularity theory provides it. In degenerate or controlled problems it can fail, which is why viscosity solutions (note 05) are needed.

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4. Worked examples (diffusion)

Example 4.1: heat equation with quadratic payoff

Take $b = 0$, $\sigma = 1$, $r = f = 0$, $g(x) = x^2$. The PDE is $\partial_t u + \frac{1}{2}\partial_{xx} u = 0$, $u(T,x) = x^2$.

Via expectation. $X_T = x + (W_T - W_t) \sim \mathcal{N}(x, \tau)$ with $\tau = T - t$, so

$$u(t,x) = \mathbb{E}[X_T^2] = x^2 + \tau. \tag{4.1}$$

Check the PDE. $\partial_t u = -1$ (since $\tau = T - t$), $\partial_{xx} u = 2$, so $-1 + \frac{1}{2}\cdot 2 = 0$ ✓. Terminal: $\tau = 0$ gives x^2 ✓.

> **Intuition:** the extra τ is the variance accumulated before maturity. A convex payoff gains value from noise — the same $\frac{1}{2}\sigma^2\partial_{xx}$ term as in Itô's lemma.

Example 4.2: Black–Scholes call

Take $dS = rS\,dt + \sigma S\,dW$ (risk-neutral GBM), constant discount rate r, $f = 0$, $g(S) = (S-K)^+$. The PDE (1.2) is the Black–Scholes equation

$$\partial_t u + rS\,\partial_S u + \tfrac{1}{2}\sigma^2 S^2 \partial_{SS} u - ru = 0, \tag{4.2}$$

and (1.3) gives $u = e^{-r\tau}\mathbb{E}[(S_T - K)^+]$. Evaluating the Gaussian integral yields the usual formula

$$u = S\,\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad d_{1,2} = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \tag{4.3}$$

where Φ is the standard normal CDF.

Numbers. $S = K = 100$, $r = 0.05$, $\sigma = 0.2$, $\tau = 1$: $d_1 = 0.35$, $d_2 = 0.15$, $u = 10.4506$. The test checks this three ways: closed form (4.3), Monte Carlo of (1.3), and a Crank–Nicolson finite-difference solve of (4.2).

Example 4.3: CARA utility of inventory (from note 01, Ex. 5.4)

Take $dS = \sigma\, dW$, fixed q and x, $g(S) = -e^{-\gamma(x+qS)}$, $r = f = 0$. Feynman–Kac says $u(t,s) = \mathbb{E}[g(S_T)]$ solves $\partial_t u + \tfrac{1}{2}\sigma^2 \partial_{ss} u = 0$.

Verify the closed form from note 01 eq. (5.5). Let

$$u(t,s) = -\exp\big(-\gamma(x+qs) + \tfrac{1}{2}\gamma^2 q^2 \sigma^2 \tau\big).$$

Then $\partial_t u = -\tfrac{1}{2}\gamma^2 q^2 \sigma^2\, u$ (the exponent decreases as t increases) and $\partial_{ss} u = \gamma^2 q^2\, u$. So $\partial_t u + \tfrac{1}{2}\sigma^2 \partial_{ss} u = u\,(-\tfrac{1}{2}\gamma^2 q^2 \sigma^2 + \tfrac{1}{2}\gamma^2 q^2 \sigma^2) = 0$ ✓.

> **Why this matters:** in note 12, Avellaneda–Stoikov guesses the value function in exactly this exponential form. The guess works because CARA turns the PDE into a problem where x and s factor out.

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5. Feynman–Kac with jumps

Setup. Add a Poisson jump as in note 01 eq. (6.2): $dX = b\,dt + \sigma\,dW + \eta\,dN$, where N has intensity λ. The generator is now (note 01 eq. 7.1)

$$\mathcal{L}\varphi(x) = b\,\partial_x \varphi + \tfrac{1}{2}\sigma^2 \partial_{xx}\varphi + \lambda\big[\varphi(x+\eta) - \varphi(x)\big]. \tag{5.1}$$

Theorem 5.1. Under analogous assumptions (with u also of polynomial growth so the jump term is integrable), Theorem 1.1 holds verbatim with this $\mathcal{L}$.

Proof sketch. Repeat §2 with note 01 eq. (6.3) instead of (5.1). The jump term $[u(X_{s-}+\eta) - u(X_{s-})]dN_s$ splits as $\lambda[\cdots]ds + [\cdots]d\tilde{N}_s$. The ds part is the new term in $\mathcal{L}$ and is cancelled by the PDE; the $d\tilde{N}$ part is a martingale increment with zero mean. □

Note the structure. The PDE is now a **nonlocal** equation: it couples $u(t,x)$ with $u(t,x+\eta)$. If X lives on a lattice (like an integer inventory), the PDE becomes a **system of ODEs**, one per lattice point. This is exactly the structure of the GLFT market-making model (note 13).

Example 5.2: Poisson counter — a system of ODEs

Take $X = N$ (pure counting process, $b = \sigma = 0$, $\eta = 1$), $g(n) = n^2$, $r = f = 0$. The state $n \in \{0, 1, 2, \dots\}$ is integer, so (1.2) becomes, for each n,

$$\frac{d}{dt}u(t,n) + \lambda\big[u(t,n+1) - u(t,n)\big] = 0, \qquad u(T,n) = n^2. \tag{5.2}$$

Via expectation. $N_T - N_t \sim \text{Poisson}(\lambda\tau)$, with mean and variance $\lambda\tau$, so

$$u(t,n) = \mathbb{E}[(n + N_T - N_t)^2] = (n+\lambda\tau)^2 + \lambda\tau. \tag{5.3}$$

Check the ODE. $\frac{d}{dt}u = -\lambda\big[2(n+\lambda\tau)+1\big]$ (chain rule with $d\tau/dt=-1$), and $\lambda[u(n+1)-u(n)] = \lambda\big[(n+1+\lambda\tau)^2-(n+\lambda\tau)^2\big] = \lambda\big[2(n+\lambda\tau)+1\big]$. Sum $=0$ ✓.

Numbers. $\lambda=2$, $\tau=1.5$, $n=3$: $u=(3+3)^2+3=39$. The test solves the truncated ODE system (5.2) numerically and compares with 39 and with Monte Carlo.

Example 5.3: expected spread income from passive quoting (market-making preview)

A market maker posts a bid and an ask, each at distance δ from mid. Assume fills on each side arrive as independent Poisson processes with intensity $\lambda(\delta)=Ae^{-k\delta}$, and each fill earns δ relative to mid. Ignoring inventory risk, the expected spread income over $[t,T]$ is (Feynman–Kac with $f=2\lambda(\delta)\delta$, $g=0$, $r=0$):

$$u(t) = 2\,Ae^{-k\delta}\,\delta\,\tau. \tag{5.4}$$

Maximizing over δ: $\frac{d}{d\delta}\big(\delta e^{-k\delta}\big) = e^{-k\delta}(1-k\delta)=0$ gives $\delta^*=1/k$.

Numbers. $k=1.5$ gives $\delta^*=0.667$. Compare with the Avellaneda–Stoikov half-spread term $\frac{1}{\gamma}\ln(1+\frac{\gamma}{k}) = 0.645$ for $\gamma=0.1$ (from the earlier chat example); as $\gamma\to 0$ it tends to $1/k$.

> **Intuition:** a risk-neutral market maker faces the monopolist's trade-off: quoting wider earns more per fill but gets fewer fills. The optimum $\delta^*=1/k$ depends only on how fast fills decay. Risk aversion ($\gamma>0$) then shrinks the spread term slightly and adds inventory skew — the content of notes 12–13.

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6. Numerical check

Script: `code/feynman_kac_checks.py`; tests: `code/tests/test_02_feynman_kac.py`.

Check	Methods compared	Expected
Heat, $g=x^2$	MC vs (4.1)	$x^2+\tau$
Black–Scholes	closed form (4.3) vs MC vs Crank–Nicolson PDE	10.4506
Poisson ODE system	ODE solve of (5.2) vs (5.3) vs MC	39
Optimal risk-neutral half-spread	grid search of (5.4) vs $1/k$	0.667

The Crank–Nicolson solver is the PDE side of Feynman–Kac, and the Monte Carlo is the expectation side; their agreement is the theorem in action.

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7. Pitfalls

1. **Time direction.** (1.2) is a *backward* equation: it has a terminal condition and is well-posed when solved from T to t. Written in $\tau=T-t$, it is a standard forward heat-type equation. The *forward* equation for the density of X is the Fokker–Planck equation (note 14), which is the adjoint of $\partial_t+\mathcal{L}$.

2. **Uniqueness needs growth conditions.** The heat equation has nonzero solutions with zero initial data that grow like e^{cx^2} (Tychonoff's example). Feynman–Kac picks the unique solution of polynomial growth; this is why (A3) is needed in Step 6.
3. u **is a function of the starting point.** $u(t,x)$ is not "the value along a path"; it is a deterministic function evaluated at the starting state. $u(s,X_s)$ along a path is what becomes a martingale.
4. **Sign of the potential.** $-ru$ in (1.2) corresponds to discounting $e^{-\int r}$. A source term $+f$ corresponds to cash received. Sign mistakes here propagate into every HJB later.

中文小结

- 定理：PDE $\partial_t u + \mathcal{L}u - ru + f = 0$，$u(T) = g$ 的解等于条件期望 $\mathbb{E}_{t,x}[D\,g(X_T) + \int D\,f\,ds]$（贴现的终值收益 + 贴现的流动收益）。这就是利率衍生品定价 PDE 的来源。
- 证明（**PDE $\Longrightarrow$ 期望**）：构造 $M_s = D\,u(s,X_s) + \int D\,f$，用 Itô 引理，PDE 恰好使漂移项为零，所以 M 是鞅，取期望即得。增长条件保证随机积分是真鞅。
- 反方向（期望 $\Longrightarrow$ **PDE**）：塔性质 + 马尔可夫性给出 $u(t,x) = \mathbb{E}[u(t+h, X_{t+h})]$（无控制的动态规划原理），再用 Dynkin 公式令 $h \to 0$ 得到 PDE。第 04 篇 HJB 推导就是在此基础上加一个 sup。
- 例子：热方程 $x^2 + \tau$；Black–Scholes 10.4506（闭式解、蒙特卡洛、Crank–Nicolson 三方一致）；CARA 指数形式满足 PDE（Avellaneda–Stoikov 猜解的基础）。
- 带跳情形：生成元多出 $\lambda[u(x+\eta) - u(x)]$，方程变为非局部；状态为整数时变成 ODE 方程组——正是 GLFT 做市模型的结构。
- 做市预览：风险中性下最优半价差 $\delta^* = 1/k$（每笔收益与成交频率的权衡），$k = 1.5$ 时为 0.667；风险厌恶会略微缩小价差并加入库存偏移。